THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 3: **World of Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following has a terminating decimal expansion? 1

(a) $\frac{3}{7}$ **(b)** $\frac{5}{18}$ **(c)** $\frac{39}{65}$ **(d)** $\frac{14}{21}$

2. The number of decimal places in the decimal expansion of $\frac{7}{400}$ is: 1

(a) 6 **(b)** 3 **(c)** 4 **(d)** 5

3. $0.\overline{12}$ in lowest terms is: 1

(a) $\frac{2}{17}$ **(b)** $\frac{1}{11}$ **(c)** $\frac{4}{33}$ **(d)** $\frac{5}{33}$

4. The length of the repeating block in the decimal expansion of $\frac{1}{7}$ is: 1

(a) 6 **(b)** 7 **(c)** 8 **(d)** 5

5. $\sqrt{3} \times \sqrt{12} =$ 1

(a) 5 **(b)** 7 **(c)** 6 **(d)** 8

6. $\sqrt{3} + (5 - \sqrt{3})$ is: 1

(a) rational **(b)** irrational **(c)** zero **(d)** negative

7. A rational number between $\frac{2}{7}$ and $\frac{4}{7}$ is: 1

(a) $\frac{6}{7}$ **(b)** $\frac{3}{7}$ **(c)** $\frac{1}{7}$ **(d)** $\frac{5}{7}$

8. $-9 + (-4) \times (-5) =$ 1

(a) 13 **(b)** 10 **(c)** 12 **(d)** 11

9. Which of the following has a terminating decimal expansion? 1

(a) $\frac{11}{39}$ **(b)** $\frac{8}{15}$ **(c)** $\frac{21}{70}$ **(d)** $\frac{4}{33}$

10. The number of decimal places in the decimal expansion of $\frac{13}{250}$ is: 1

(a) 4 **(b)** 2 **(c)** 3 **(d)** 5

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Express $0.8\overline{3}$ as a fraction. 2

12. Find the decimal expansion of $\frac{5}{22}$. 2

13. Simplify $(8 + \sqrt{5})(8 - \sqrt{5})$. 2

14. Express $0.6\overline{1}$ as a fraction. 2

MASTERMIND

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Express $0.\overline{027}$ as a fraction in lowest terms. 3

16. Express $1.2\overline{3}$ in the form $\frac{p}{q}$. 3

17. Without dividing, state whether $\frac{21}{75}$ has a terminating decimal expansion. 3

OR

Express $0.\overline{135}$ as a fraction in lowest terms. 3

18. Express $2.4\overline{6}$ in the form $\frac{p}{q}$. 3

OR

Without dividing, state whether $\frac{13}{80}$ has a terminating decimal expansion. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Express $1.1\overline{6}$ as $\frac{p}{q}$ and verify by long division of p by q. 5

OR

Simplify $(3+\sqrt{5})^2+(3-\sqrt{5})^2$ and state whether the result is rational. 5

20. Express $2.3\overline{3}$ as $\frac{p}{q}$ and verify by long division of p by q. 5

OR

Simplify $(4+\sqrt{3})^2+(4-\sqrt{3})^2$ and state whether the result is rational. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND